

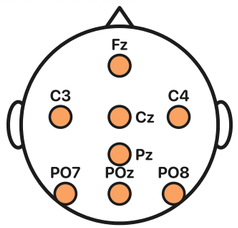
Decoding dance moves from EEG

From cued dance sequences to live, move-by-move decoding with an 8-channel EEG headset

DATA cued dance sequences, labelled EEG
MODEL windows → features → random forest
LIVE sliding-window decoding, avatar feedback

1 Collect — DATA

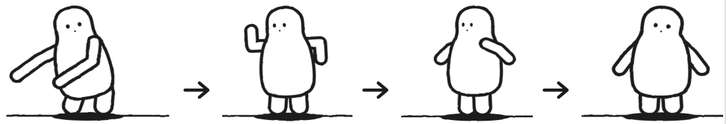
Wear the EEG headset



g.tec BCI Core-8
8 dry electrodes
sampled at **250 Hz**
streamed over LSL,
raw & unfiltered

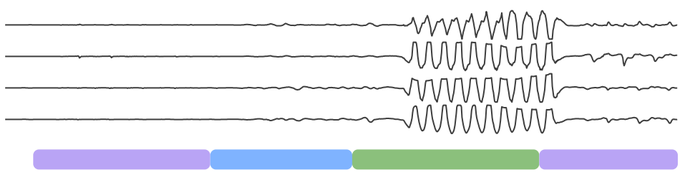
Movement artifacts and muscle activity are kept: they carry the move.

Dance cued sequences



Moves chained back to back for 3–6 s each, with a 1 s preview of the next move. 5 sequences per block, 10 s breaks.

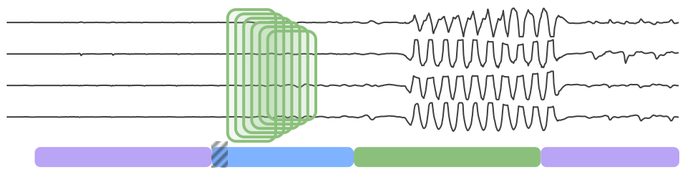
Label every sample



Each cue sends a marker into the stream, so every sample gets its move (real recording; 4 of 8 channels, 21 s).

2 Train — MODEL

Cut into sliding windows



1.5 s windows every 0.25 s, kept when fully inside one move. The first 0.5 s after each cue (hatched) is reaction time.

Describe each window: 92 features

per channel (8 × 8 = 64)

log RMS · log RMS of Δ · motion 0.5–3 Hz · θ 3–8 · α 8–13 · β 13–30 · γ 30–45 · EMG 55–95

\+ **channel pairs** 28 correlations

Only the mean is removed: slow motion artifacts and high-frequency EMG are the signal. 50 Hz mains is left out of the bands.

Fit the classifier

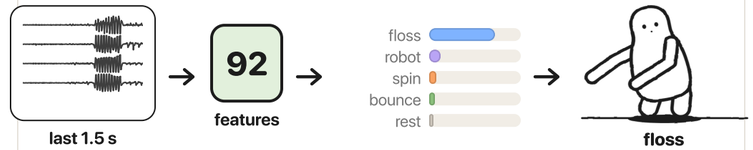


Random forest
300 trees, class-balanced
Leave-one-block-out
tested on blocks never
seen in training

One model per participant, trained on all recorded blocks.

3 Decode — LIVE

Classify every 0.25 s



The same windows and features as in training, on the live stream.

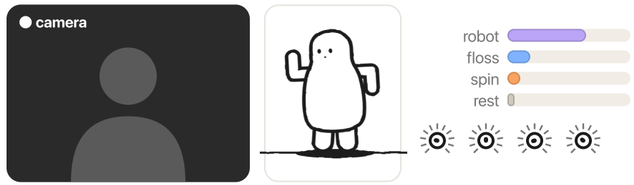
Choose how fast to switch

Instant
winner takes all:
top move shown at once

Steady
smoothing, threshold
& hold: fewer flickers

Knobs trade latency for stability. On unseen blocks, Instant catches 92% of moves with 1.7 s median latency.

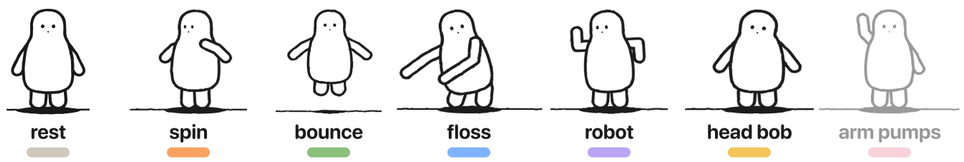
Show the dance back



Demo dashboard: camera, the decoded move as an avatar, probabilities and live knobs.

Move library

avatars used in the cues and the live display (arm pumps: seq2 only)



Dataset · participant p01

seq2	7 moves · 2 blocks	**704 windows**
seq3	6 moves · 2 blocks	**582 windows**